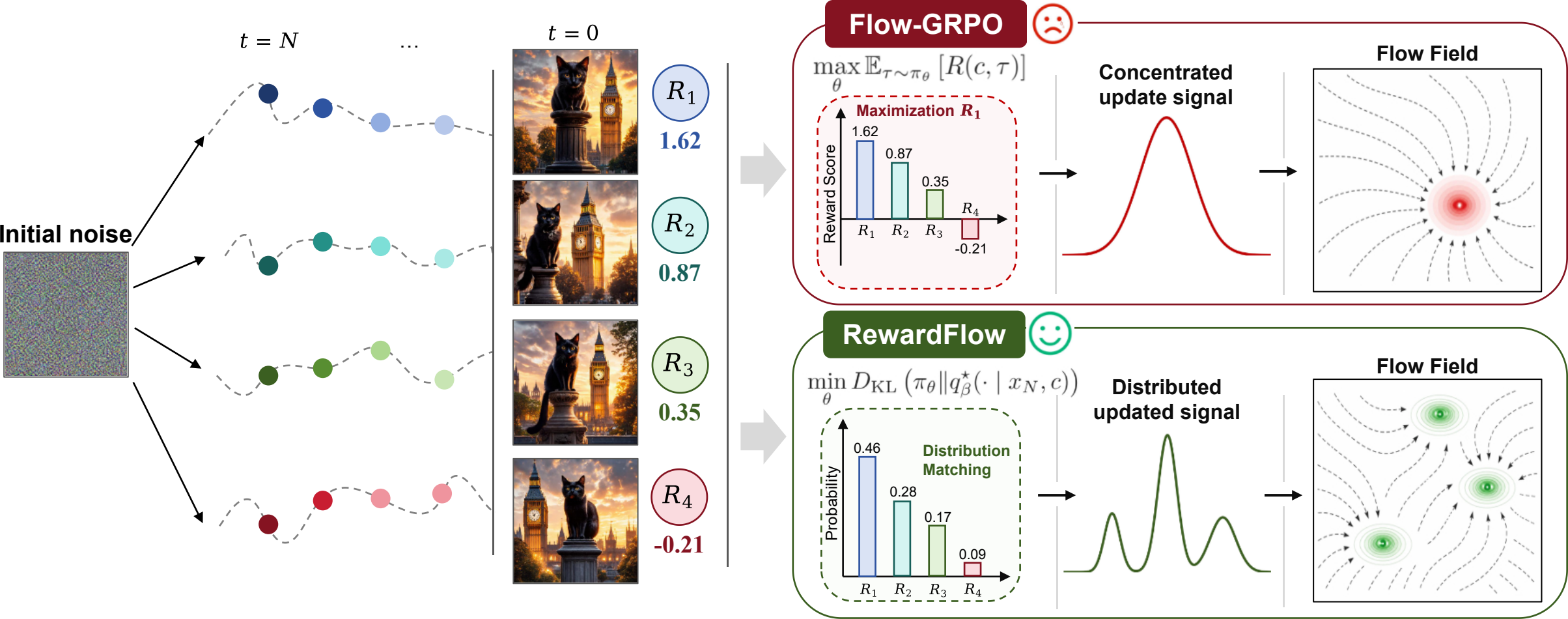

$t = N$
...
$t = 0$
Initial noise
R_1
1.62
R_2
0.87
R_3
0.35
R_4
-0.21
Flow-GRPO
$\max_\theta \mathbb{E}_{\tau\sim\pi_\theta}[R(c,\tau)]$
Maximization R_1
Reward Score
1.62
0.87
0.35
-0.21
R_1 R_2 R_3 R_4
Concentrated update signal
Flow Field
RewardFlow
$\min_\theta D_{\mathrm{KL}}\left(\pi_\theta \| q^\star_\beta(\cdot \mid x_N, c)\right)$
Probability
0.46
0.28
0.17
0.09
Distribution Matching
R_1 R_2 R_3 R_4
Distributed updated signal
Flow Field